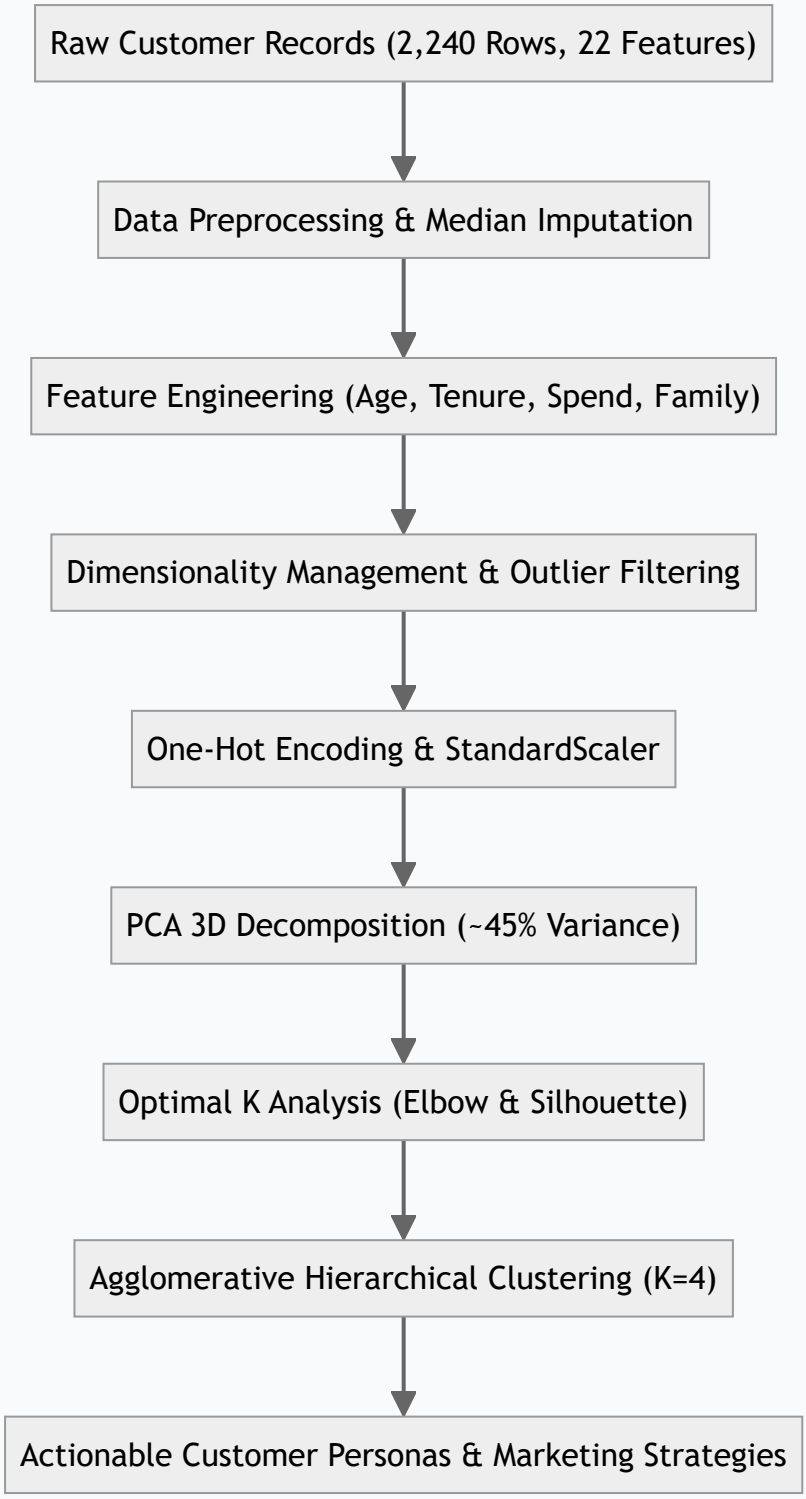
Submodule 33.01: Project Introduction & Business Objectives

Topic Overview & Business Context

In modern retail and e-commerce platforms, customer segmentation is one of the most commercially critical applications of **Unsupervised Machine Learning**. Rather than treating all customers uniformly with generic, mass marketing campaigns ("spray-and-pray"), businesses must group customers into cohesive behavioral clusters. This enables personalized product recommendations, optimized marketing budget allocation, and elevated customer retention rates.

In this project, we engineer the **SmartCart Clustering System** for an international e-commerce company serving customers across multiple regions. The dataset comprises 2,240 customer transaction records across 22 multi-faceted attributes.



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1. The Business Problem & Objectives

- **Current Operational Bottleneck**: SmartCart currently executes generic marketing and promotional strategies.
- **Core Deficiencies**:

    - Inability to distinguish high-value loyal spenders from discount-seeking casual visitors.
    - Suboptimal return on marketing investment (marketing spend wasted on unengaged leads).
    - High customer churn due to irrelevant promotional bombardment.
- **Project Goal**: Build an intelligent customer segmentation pipeline using unsupervised clustering algorithms to discover hidden behavioral archetypes and empower data-driven marketing decisions.
- **Target Output**: Segments mapped to distinct customer personas with customized business recommendations.

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2. Dataset Architecture & Attribute Categorization

The dataset (`smartcart_customers.csv`) contains 2,240 customer records across 22 attributes:

Category	Attributes	Business Significance
Demographics	`ID` , `Year_Birth` , `Education` , `Marital_Status` , `Income` , `Kidhome` , `Teenhome`	Personal profile, earning capacity, and household dependency structure.
Product Expenditures	`MntWines` , `MntFruits` , `MntMeatProducts` , `MntFishProducts` , `MntSweetProducts` , `MntGoldProds`	Basket composition and category preferences over the last 2 years.
Purchasing Channels	`NumDealsPurchases` , `NumWebPurchases` , `NumCatalogPurchases` , `NumStorePurchases`	Channel preferences (e-commerce web vs brick-and-mortar store vs mail catalog).
Engagement & Recency	`Dt_Customer` , `Recency` , `NumWebVisitsMonth` , `Complain` , `Response`	Customer loyalty, days since last purchase, website activity, friction, and campaign response.

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3. Data Ingestion & Initial Exploratory Audit

The official project code is located at:

- smartcart.ipynb
- smartcart_customers.csv

Python Implementation (Notebook Cells 00 - 04)

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load dataset
df = pd.read_csv("smartcart_customers.csv")

# Display initial rows
print(df.head())

# Check dataset dimensions
print("Dataset Shape:", df.shape) # Output: (2240, 22)

# Audit missing values across columns
print(df.isnull().sum())
```

Initial Health Check Findings:

1. **Dimensions**: 2,240 rows and 22 feature columns.
2. **Missing Values**:
    - `Income` : Exactly 24 null values present.
    - All other 21 columns have 0 missing values.
3. **No Target Label**: There is no ground-truth target vector y. All features represent input attributes X for exploratory unsupervised manifold learning.